

V2: Economic Logic & Context Framework

(Audit-Ready)

Integrating Tiered Context, Model Escalation, and Multi-Tenant Safety Guardrails

1. The Microscope-to-Telescope Context Model

This model optimizes TTFT (Time-To-First-Token) and token expenditure by dynamically expanding the agent's field of vision only upon request.

View	Context Payload	Access Mechanism
Microscope	Today's Recovery (HRV/Sleep) + Planned Session	Loaded on <code>init()</code>
Magnifying Glass	7-Day Performance Trends + Volume Accumulation	Skill: <code>get_weekly_stats()</code>
Telescope	Macrocycle Progress + Longitudinal Limiters	Skill: <code>query_long_term_trends()</code>

2. Production Fixes & Architectural Critiques

FIX 1: Dynamic Directive Injection (The Supervisor Loop)

Problem: Global audit updates were stored but not ingested by active agents.

Solution: The `init()` function now performs a parallel fetch of the static Role and the dynamic `directives_update.md`. This creates a self-correcting prompt at every session start.

FIX 2: Stateful Model Escalation (Context Handoff)

Problem: Upgrading from Haiku to Sonnet caused conversational amnesia.

Solution: The `upgrade_model` Skill serializes the current `messages[]` array and passes it as the `initial_state` for the new high-reasoning session.

FIX 3: Graceful Degradation (Usage Quotas)

Problem: "Power users" burning flat-rate margins.

Solution: Implement a D1-backed Feature Flag. If credits hit zero, the Agent interface is locked, and the app reverts to a high-performance manual logging UI via standard Kysely endpoints.

3. Implementation: Stateful Escalation Logic

```
// Skill: trigger_high_reasoning.ts export async function run(currentSession) { const
history = currentSession.getMessages(); // Re-initialize Flue with higher-tier model await
init({ model: 'anthropic/claude-3-5-sonnet', sandbox: currentSession.sandbox,
initialMessages: history, // Prevent context drop system: await fetchLayeredDirectives()
// Ingest Supervisor updates }); return "Model upgraded. Analyzing biomechanics with high
precision..."; }
```

4. Audit Loop & Financial Controls

The Supervisor (Sonnet) audits samples of low-cost (Haiku) interactions weekly. Findings are pushed to the global directives file, closing the loop on quality without manual engineering intervention.

- **Audit Frequency:** 5% of all successful interactions.
- **Cost Protection:** UsageQuotas table updated via `context.waitUntil` to ensure zero-latency tracking.